Demo document — not a scientific publication

MotionTiny: distilled motion features for streaming video

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Abstract
MotionTiny distils a large teacher network into a compact student for streaming action recognition. Training uses the DemoKinetics-200 training split together with unlabelled clips from DemoStream-1M, so supervision is partly self-distilled rather than fully supervised.

2. Method

The student uses a ResNet-18 backbone with a temporal shift module. We sample 8 frames per clip at 168x168 pixels.

4. Results

MotionTiny reaches 79.2% top-1 accuracy on the ED-Bench test split at 8 frames per clip. For reference, the teacher network reaches 84.0% top-1 on the same split but is far too large for embedded deployment.

We report 12 ms per clip on an NVIDIA A100 GPU. We did not measure latency on embedded hardware.

5. Limitations

Because the reported latency comes from a datacentre GPU, it is not comparable with numbers measured on embedded devices.